

Architecting reproducible science: A practical Path Beyond the Notebook

Software Engineering principles to improve your HPC

Fernando Garzon, March 2026

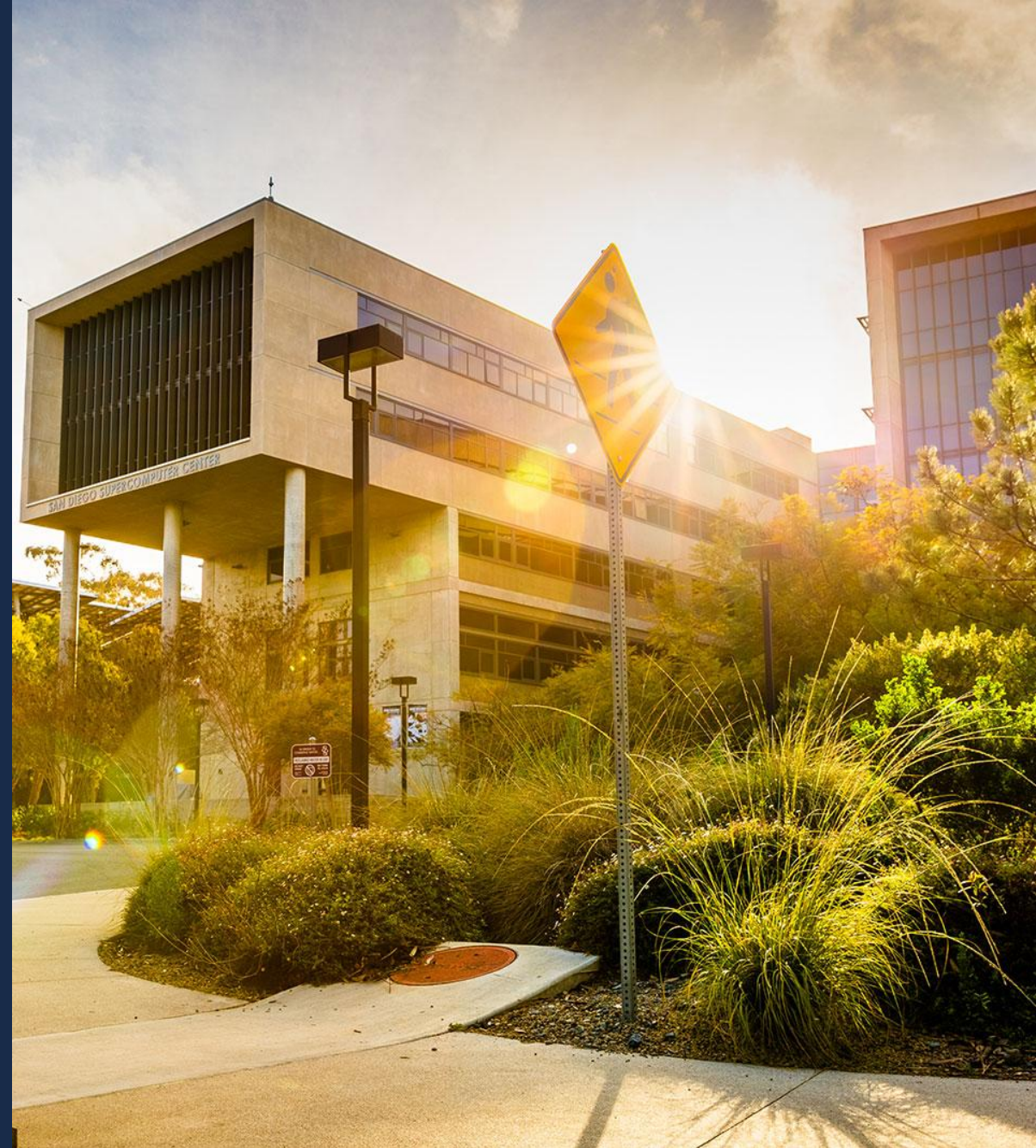
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SDSC
SAN DIEGO SUPERCOMPUTER CENTER

UC San Diego

Who's Fernando Garzon

- Software Engineer
- Computational & Data Scientist Research Specialist at SDSC
- Work on User Support for TSCC.
- Also work in Software Dev (SWD) projects within SDSC.
- Passionate about SWD, DevOps, Automation, Cloud Architecture, IaC and more.
- Automation Enthusiast.

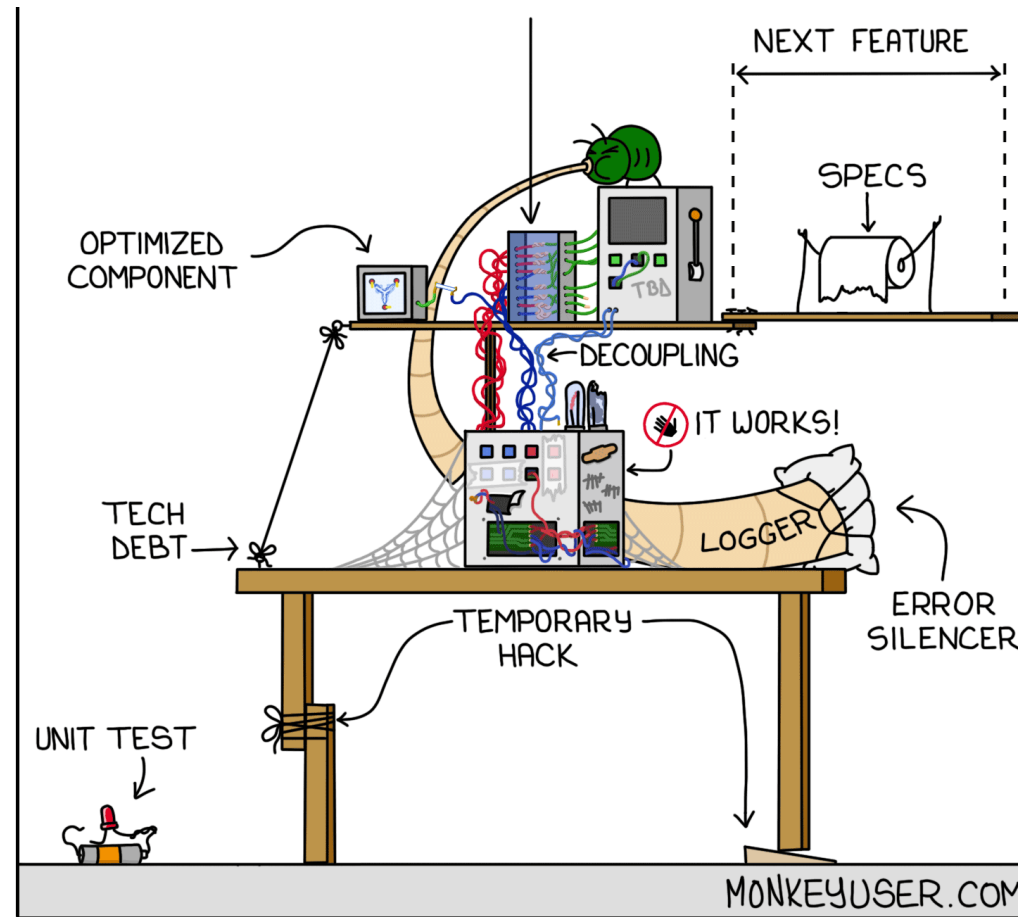




Outline

1. Intro: The un-reproducibility horror
2. Notebooks Don't Scale. Packages Do.
3. Let's Prod this.
4. Lights, Camera, GitHub Actions!
5. MLOps – Turning your Notebook into a Money-Printing Machine.
6. Final remarks

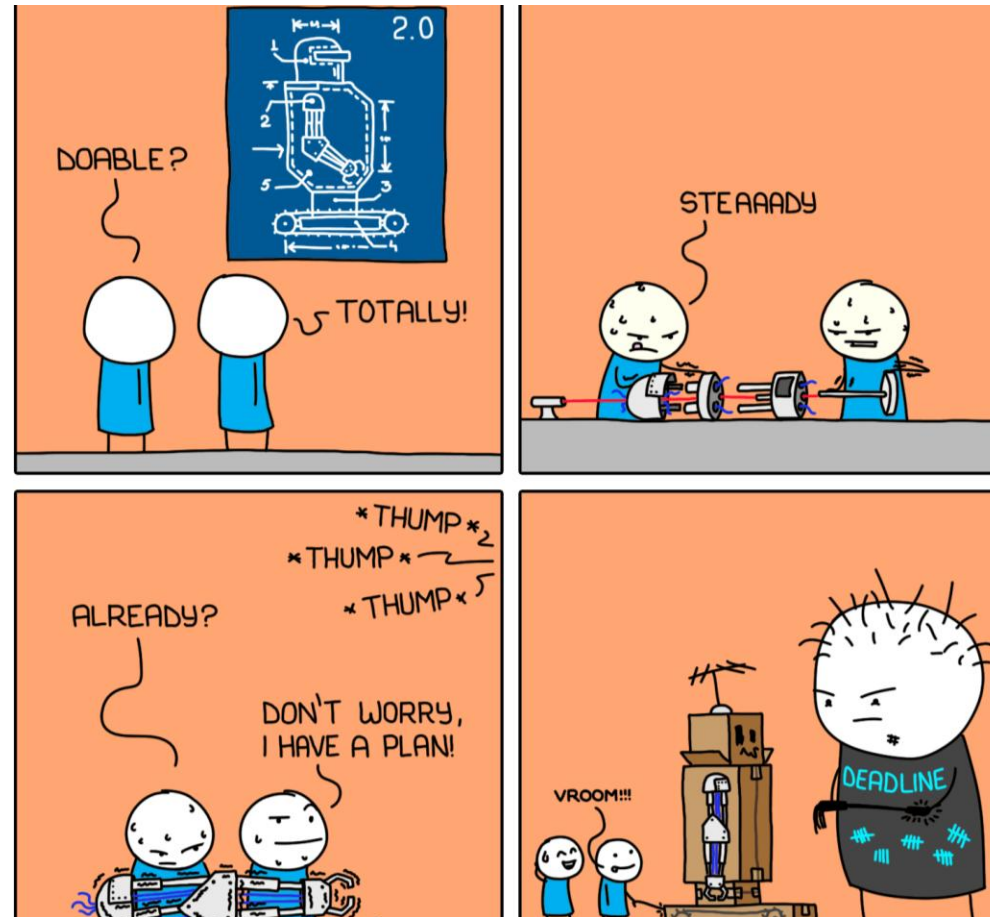
Intro: The un-reproducibility horror



Intro video:

[Link - Video](#)

Notebooks Don't Scale. Packages Do.

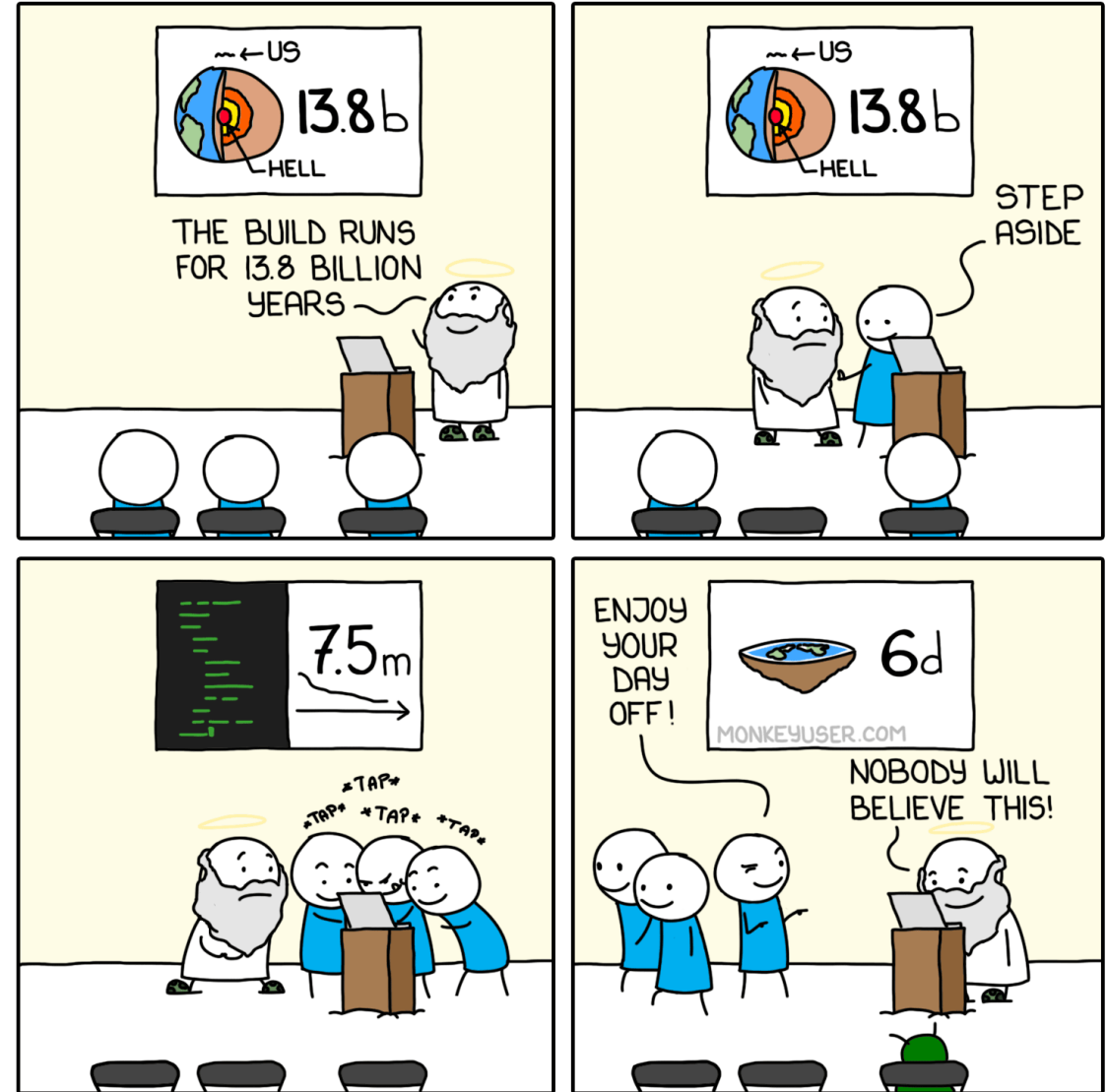


Let's Prod this

Follow the instructions in here:

https://github.com/ofgarzon2662/mnist_ae

NEW WORLD



Lights, Camera, GitHub Actions!

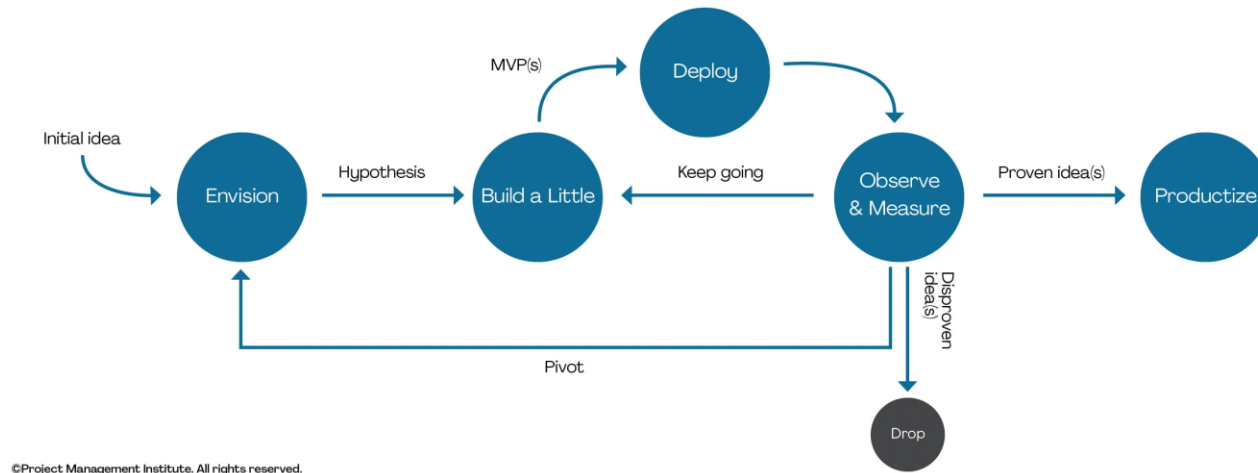
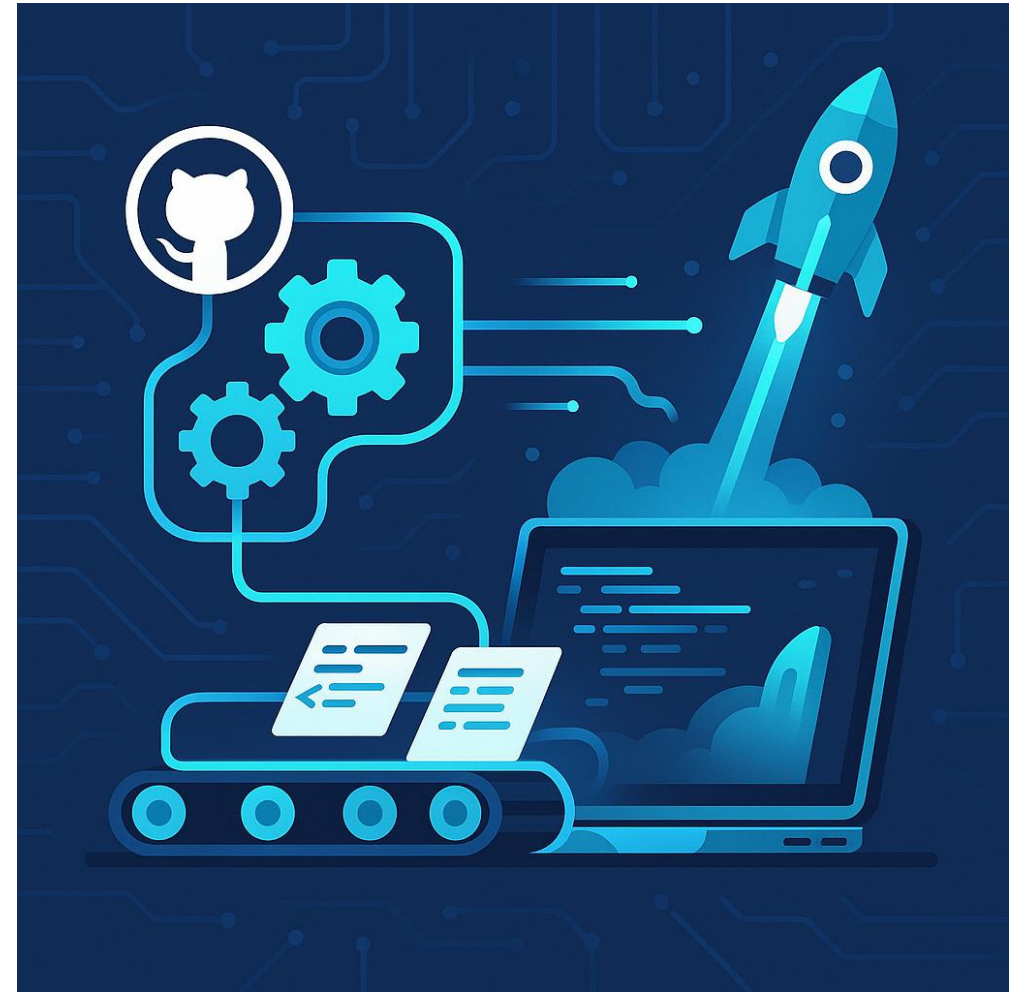
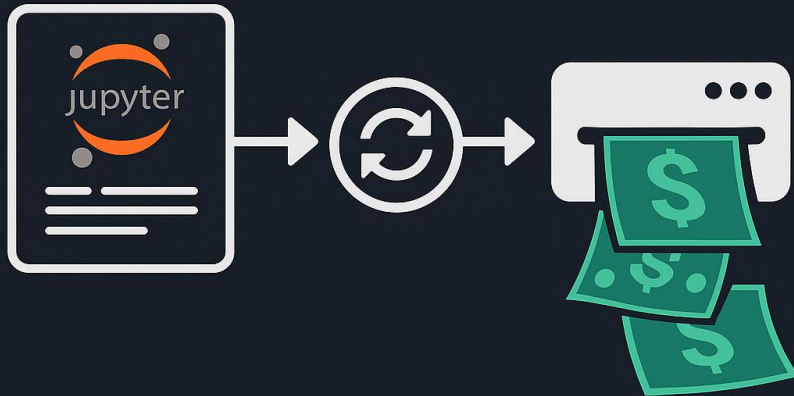


Image reference: <https://www.pmi.org/disciplined-agile/lifecycle/exploratory-lifecycle>



MLOps: Turning your Notebook into a Money-Printing Machine.

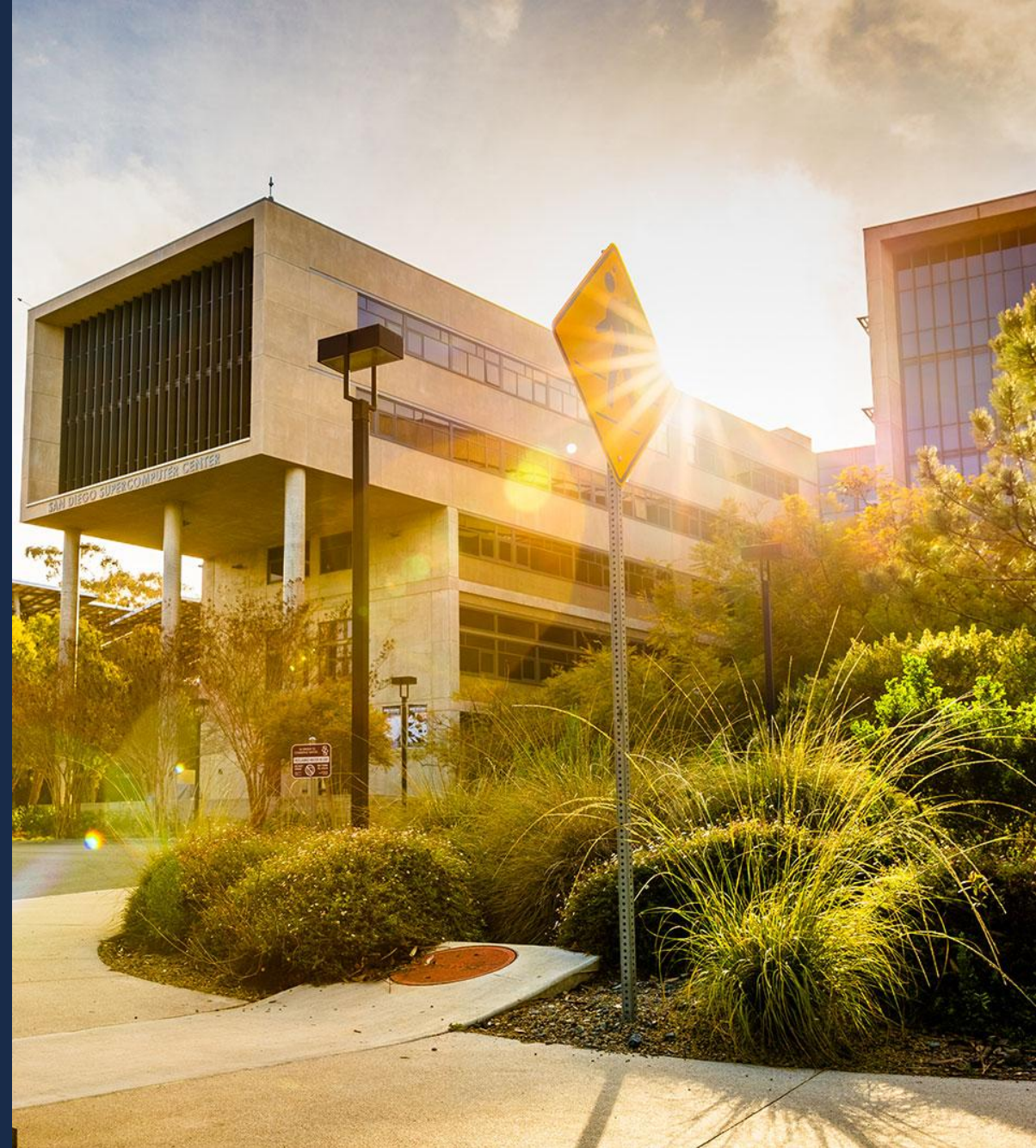
MLOps: Turning your Notebook into a Money-Printing Machine



1. Amazon: 136k Deps. a day.
2. Google: $O(10k)$ Deps a day.
3. Netflix: 500 Deps. a day.
4. Etsy: 50 Deps. a day.
5. Meta: 1 Dep. a day.

Final Remarks

1. There are bad ML/AI Engineers, and then there are ML/AI Engineers who can deploy their code to prod.
2. Don't commit big resources and time into testing ML/AI models. Use automated Unit Test first.
3. Github Actions are the 1st step in the automation process but there is plenty more you can learn to leverage your computing practices: (CI/CD, DevOps, MLOps).
4. AI won't take your AI Engineering job. Someone who knows how to use DevOps pipelines to streamline the deployment of AI/ML models into production and generates absurd amounts of revenue, that person will.





Questions



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